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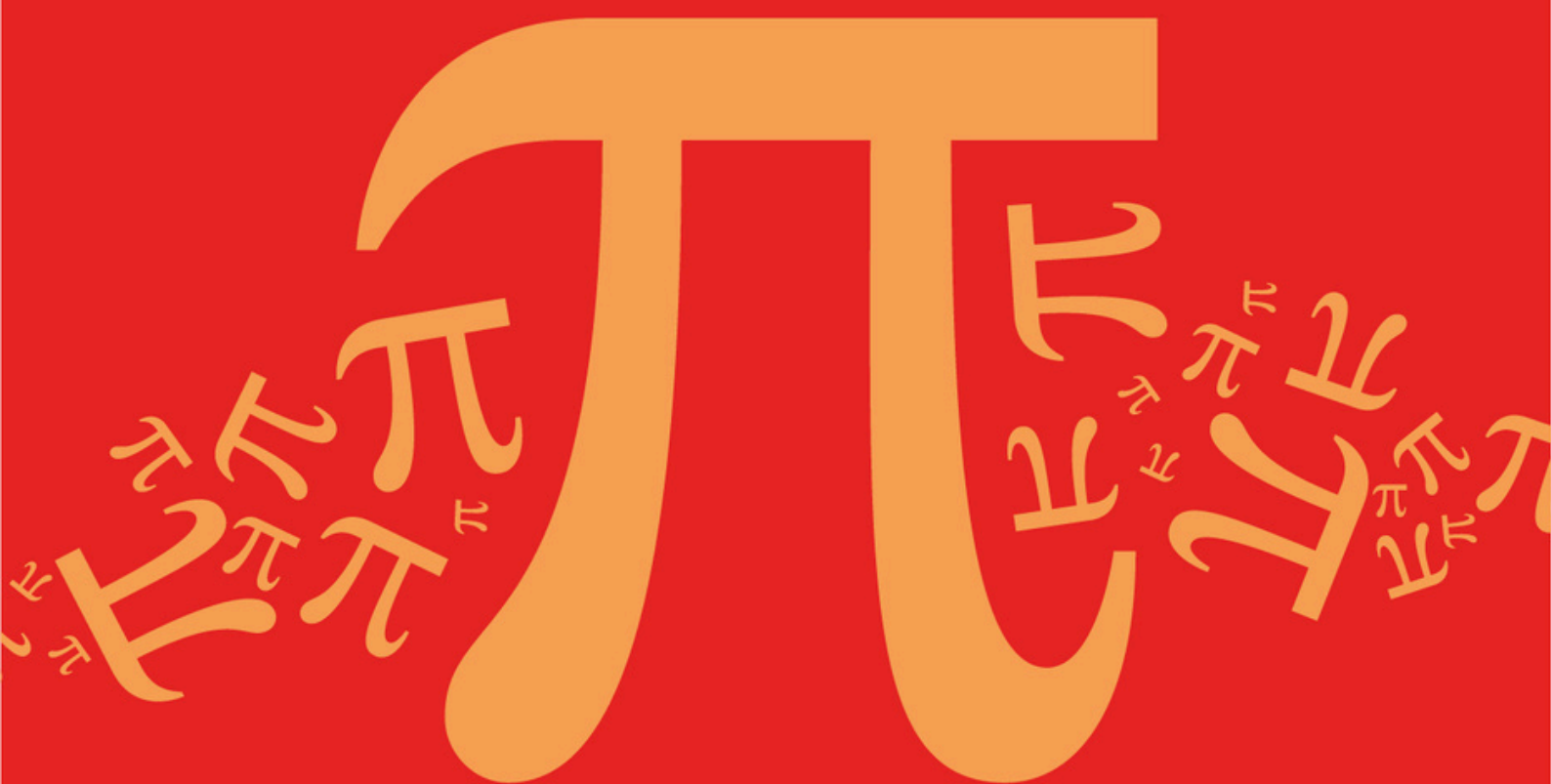
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AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D1: Sums and Products of Roots

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic **D: Further Algebra and Functions**
Sub topic **D1: Sums and Products of Roots**

EXAM PAPERS PRACTICE

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

α, β and γ are the real roots of the cubic equation

$$x^3 + mx^2 + nx + 2 = 0$$

By considering $(\alpha - \beta)^2 + (\gamma - \alpha)^2 + (\beta - \gamma)^2$, prove that

$$m^2 \geq 3n$$

(Total 4 marks)

Q2.

The equation $x^3 - 8x^2 + cx + d = 0$, where c and d are real numbers, has roots α, β, γ .

When plotted on an Argand diagram, the triangle with vertices at α, β, γ has an area of 8.

Given $\alpha = 2$, find the values of c and d .

Fully justify your solution.

(Total 5 marks)

Q3.

The roots of the quadratic equation

$$x^2 + 2x - 5 = 0$$

are α and β

(a) Write down the value of $\alpha + \beta$ and the value of $\alpha\beta$.

(2)

(b) Calculate the value of $\alpha^2 + \beta^2$.

(2)

(c) Find a quadratic equation which has roots $\alpha^3\beta + 1$ and $\alpha\beta^3 + 1$.

(5)

(Total 9 marks)

Q4.

The roots of the equation

$$z^3 - 5z^2 + kz - 4 = 0$$

are α, β and γ .

(a) (i) Write down the value of $\alpha + \beta + \gamma$ and the value of $\alpha\beta\gamma$.

(2)

(ii) Hence find the value of $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$.

(2)

(b) The value of $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$ is -4 .

(i) Explain why α , β and γ cannot all be real.

(1)

(ii) By considering $(\alpha\beta + \beta\gamma + \gamma\alpha)^2$, find the possible values of k .

(4)

(Total 9 marks)

Q5.

The equation

$$2x^2 + 3x - 6 = 0$$

has roots α and β .

(a) Write down the value of $\alpha + \beta$ and the value of $\alpha\beta$.

(2)

(b) Hence show that $\alpha^3 + \beta^3 = -\frac{135}{8}$.

(3)

(c) Find a quadratic equation, with integer coefficients, whose roots are $\alpha + \frac{\alpha}{\beta^2}$ and $\beta + \frac{\beta}{\alpha^2}$.

(6)

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Q6.

The cubic equation

$$z^3 + pz^2 + qz + 37 - 36i = 0$$

where p and q are constants, has three complex roots, α , β and γ . It is given that $\beta = -2 + 3i$ and $\gamma = 1 + 2i$.

(a) (i) Write down the value of $\alpha\beta\gamma$.

(1)

(ii) Hence show that $(8 + i)\alpha = 37 - 36i$.

(2)

(iii) Hence find α , giving your answer in the form $m + ni$, where m and n are integers.

(3)

(b) Find the value of p .

(1)

- (c) Find the value of the complex number q .

(2)

(Total 9 marks)

Q7.

The quadratic equation

$$2x^2 + 7x + 8 = 0$$

has roots α and β .

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (b) Show that $\alpha^2 + \beta^2 = \frac{17}{4}$.

(2)

- (c) Find a quadratic equation, with integer coefficients, which has roots

$$\frac{1}{\alpha^2} \text{ and } \frac{1}{\beta^2}$$

(5)

(Total 9 marks)

Q8.

The numbers α , β and γ satisfy the equations

$$\alpha^2 + \beta^2 + \gamma^2 = -10 - 12i$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = 5 + 6i$$

- (a) Show that $\alpha + \beta + \gamma = 0$.

(2)

- (b) The numbers α , β and γ are also the roots of the equation

$$z^3 + pz^2 + qz + r = 0$$

Write down the value of p and the value of q .

(2)

- (c) It is also given that $\alpha = 3i$.

- (i) Find the value of r .

(3)

- (ii) Show that β and γ are the roots of the equation

$$z^2 + 3iz - 4 + 6i = 0$$

(2)

- (iii) Given that β is real, find the values of β and γ .

(3)

(Total 12 marks)

Q9.

The quadratic equation

$$5x^2 - 7x + 1 = 0$$

has roots α and β .

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (b) Show that $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{39}{5}$.

(3)

- (c) Find a quadratic equation, with integer coefficients, which has roots

$$\alpha + \frac{1}{\alpha} \text{ and } \beta + \frac{1}{\beta}$$

(5)

(Total 10 marks)

Q10.

The cubic equation

$$z^3 + pz + q = 0$$

has roots α , β and γ .

- (a) (i) Write down the value of $\alpha + \beta + \gamma$.

(1)

- (ii) Express $\alpha\beta\gamma$ in terms of q .

(1)

- (b) Show that

$$\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$$

(3)

- (c) Given that $\alpha = 4 + 7i$ and that p and q are real, find the values of :

- (i) β and γ ;

(2)

- (ii) p and q .

(3)

- (d) Find a cubic equation with integer coefficients which has roots, $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$.

(3)

(Total 13 marks)

Q11.

The quadratic equation $x^2 - 6x + 18 = 0$ has roots α and β .

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (b) Find a quadratic equation, with integer coefficients, which has roots α^2 and β^2 .

(4)

- (c) Hence find the values of α^2 and β^2 .

(1)

(Total 7 marks)

Q12.

- (a) Show that $(1 + i)^3 = 2i - 2$.

(2)

- (b) The cubic equation

$$z^3 - (5 + i)z^2 + (9 + 4i)z + k(1 + i) = 0$$

where k is a real constant, has roots α , β and γ .

It is given that $\alpha = 1 + i$.

- (i) Find the value of k .

(3)

- (ii) Show that $\beta + \gamma = 4$.

(1)

- (iii) Find the values of β and γ .

(5)

(Total 11 marks)

Q13.

The equation

$$4x^2 + 6x + 3 = 0$$

has roots α and β .

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

(b) Show that $\alpha^2 + \beta^2 = \frac{3}{4}$.

(2)

(c) Find an equation, with integer coefficients, which has roots

$$3a - \beta \text{ and } 3\beta - a$$

(5)

(Total 9 marks)

Q14.

The cubic equation

$$z^3 - 2z^2 + k = 0 \quad (k \neq 0)$$

has roots α , β and γ .

(a) (i) Write down the values of $\alpha + \beta + \gamma$ and $\alpha\beta + \beta\gamma + \gamma\alpha$.

(2)

(ii) Show that $\alpha^2 + \beta^2 + \gamma^2 = 4$.

(2)

(iii) Explain why $\alpha^3 - 2\alpha^2 + k = 0$.

(1)

(iv) Show that $\alpha^3 + \beta^3 + \gamma^3 = 8 - 3k$.

(2)

(b) Given that $\alpha^4 + \beta^4 + \gamma^4 = 0$:

(i) show that $k = 2$;

(4)

(ii) find the value of $\alpha^5 + \beta^5 + \gamma^5$.

(3)

(Total 14 marks)

Q15.

The quadratic equation

$$3x^2 - 6x + 1 = 0$$

has roots α and β .

(a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

(b) Show that $\alpha^3 + \beta^3 = 6$.

(3)

- (c) Find a quadratic equation, with integer coefficients, which has roots $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$.

(4)

(Total 9 marks)

Q16.

The cubic equation

$$2z^3 + pz^2 + qz + 16 = 0$$

where p and q are real, has roots α , β and γ .

It is given that $\alpha = 2 + 2\sqrt{3}i$.

- (a) (i) Write down another root, β , of the equation.

(1)

- (ii) Find the third root, γ .

(3)

- (iii) Find the values of p and q .

(3)

- (b) (i) Express α in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(2)

- (ii) Show that

$$(2 + 2\sqrt{3}i)^n = 4^n \left(\cos \frac{2n\pi}{3} + i \sin \frac{2n\pi}{3} \right)$$

(2)

- (iii) Show that

$$\alpha^n + \beta^n + \gamma^n = 2^{2n+1} \cos \frac{2n\pi}{3} + \left(-\frac{1}{2} \right)^n$$

where n is an integer.

(3)

(Total 14 marks)

Q17.

The quadratic equation

$$x^2 - 4x + 10 = 0$$

has roots α and β .

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

(b) Show that $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{2}{5}$.

(2)

(c) Find a quadratic equation, with integer coefficients, which has roots $\alpha + \frac{2}{\beta}$ and $\beta + \frac{2}{\alpha}$.

(6)

(Total 10 marks)

Q18.

The roots of the cubic equation

$$z^3 - 2z^2 + pz + 10 = 0$$

are α , β and γ .

It is given that $\alpha^3 + \beta^3 + \gamma^3 = -4$.

(a) Write down the value of $\alpha + \beta + \gamma$.

(1)

(b) (i) Explain why $\alpha^3 - 2\alpha^2 + p\alpha + 10 = 0$.

(1)

(ii) Hence show that

$$\alpha^2 + \beta^2 + \gamma^2 = p + 13$$

(4)

(iii) Deduce that $p = -3$.

(2)

(c) (i) Find the real root α of the cubic equation $z^3 - 2z^2 - 3z + 10 = 0$.

(2)

(ii) Find the values of β and γ .

(3)

(Total 13 marks)

Q19.

It is given that α , β and γ satisfy the equations

$$\alpha + \beta + \gamma = 1$$

$$\alpha^2 + \beta^2 + \gamma^2 = -5$$

$$\alpha^3 + \beta^3 + \gamma^3 = -23$$

- (a) Show that $\alpha\beta + \beta\gamma + \gamma\alpha = 3$.

(3)

- (b) Use the identity

$$(\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \alpha\beta - \beta\gamma - \gamma\alpha) = \alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma$$

to find the value of $\alpha\beta\gamma$.

(2)

- (c) Write down a cubic equation, with integer coefficients, whose roots are α , β and γ .

(2)

- (d) Explain why this cubic equation has two non-real roots.

(2)

- (e) Given that α is real, find the values of α , β and γ .

(4)

(Total 13 marks)

Q20.

The equation

$$2x^2 + x - 8 = 0$$

has roots α and β .

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (b) Find the value of $\alpha^2 + \beta^2$.

(2)

- (c) Find a quadratic equation which has roots $4\alpha^2$ and $4\beta^2$. Give your answer in the form $x^2 + px + q = 0$, where p and q are integers.

(3)

(Total 7 marks)

Q21.

The cubic equation

$$z^3 + pz^2 + 25z + q = 0$$

where p and q are real, has a root $\alpha = 2 - 3i$.

- (a) Write down another non-real root, β , of this equation.

(1)

- (b) Find:

- (i) the value of $\alpha\beta$; (1)
- (ii) the third root, γ , of the equation; (3)
- (iii) the values of p and q . (3)
- (Total 8 marks)

Q22.

- (a) (i) It is given that α and β are the roots of the equation

$$x^2 - 2x + 4 = 0$$

Without solving this equation, show that α^3 and β^3 are the roots of the equation

$$x^2 + 16x + 64 = 0$$

- (6)

- (ii) State, giving a reason, whether the roots of the equation

$$x^2 + 16x + 64 = 0$$

are real and equal, real and distinct, or non-real.

(2)

- (b) Solve the equation

$$x^2 - 2x + 4 = 0$$

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(2)

- (c) Use your answers to parts (a) and (b) to show that

$$(1 + i\sqrt{3})^3 = (1 - i\sqrt{3})^3$$

(2)

(Total 12 marks)

Q23.

The cubic equation

$$z^3 + iz^2 + 3z - (1 + i) = 0$$

has roots α , β and γ .

- (a) Write down the value of:

(i) $\alpha + \beta + \gamma$;

(1)

(ii) $\alpha\beta + \beta\gamma + \gamma\alpha$;

(1)

(iii) $\alpha\beta\gamma$.

(1)

(b) Find the value of:

(i) $\alpha^2 + \beta^2 + \gamma^2$;

(3)

(ii) $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$;

(4)

(iii) $\alpha^2\beta^2\gamma^2$.

(2)

(c) Hence write down a cubic equation whose roots are α^2 , β^2 and γ^2 .

(2)

(Total 14 marks)

Q24.

The equation

$$x^2 + x + 5 = 0$$

has roots α and β .

(a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

(b) Find the value of $\alpha^2 + \beta^2$.

(2)

(c) Show that $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = -\frac{9}{5}$.

(2)

(d) Find a quadratic equation, with integer coefficients, which has roots $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$

(2)

(Total 8 marks)

Q25.

The cubic equation

$$z^3 + qz + (18 - 12i) = 0$$

where q is a complex number, has roots α , β and γ .

(a) Write down the value of:

(i) $\alpha\beta\gamma$;

- (1)
- (ii) $\alpha + \beta + \gamma$. (1)
- (b) Given that $\beta + \gamma = 2$, find the value of:
- (i) α ; (1)
- (ii) $\beta\gamma$; (2)
- (iii) q . (3)
- (c) Given that β is of the form ki , where k is real, find β and γ . (4)
- (Total 12 marks)

Q26.

The quadratic equation

$$2x^2 + 4x + 3 = 0$$

has roots α and β

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$. (2)
- (b) Show that $\alpha^2 + \beta^2 = 1$. (3)
- (c) Find the value of $\alpha^4 + \beta^4$. (3)
- (Total 8 marks)

Q27.

The cubic equation

$$z^3 + 2(1 - i)z^2 + 32(1 + i) = 0$$

has roots α , β and γ .

- (a) It is given that α is of the form ki , where k is real. By substituting $z = ki$ into the equation, show that $k = 4$. (5)
- (b) Given that $\beta = -4$, find the value of γ . (2)
- (Total 7 marks)

Q28.

The quadratic equation

$$2x^2 - x + 4 = 0$$

has roots α and β .

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (b) Show that $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{1}{4}$.

(2)

- (c) Find a quadratic equation with integer coefficients such that the roots of the equation are $\frac{4}{\alpha}$ and $\frac{4}{\beta}$.

(3)

(Total 7 marks)

Q29.

The cubic equation

$$z^3 + pz^2 + 6z + q = 0$$

has roots α , β , and γ .

- (a) Write down the value of $\alpha\beta + \beta\gamma + \gamma\alpha$.

(1)

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- (b) Given that p and q are real and that $\alpha^2 + \beta^2 + \gamma^2 = -12$:

- (i) explain why the cubic equation has two non-real roots and one real root;

(2)

- (ii) find the value of p .

(4)

- (c) One root of the cubic equation is $-1 + 3i$.

Find:

- (i) the other two roots;

(3)

- (ii) the value of q .

(2)

(Total 12 marks)

Q30.

- (a) (i) Calculate $(2 + i\sqrt{5})(\sqrt{5} - i)$. (3)

- (ii) Hence verify that $\sqrt{5} - i$ is a root of the equation

$$(2 + i\sqrt{5})z = 3z^*$$

where z^* is the conjugate of z .

(2)

- (b) The quadratic equation

$$x^2 + px + q = 0$$

in which the coefficients p and q are real, has a complex root $\sqrt{5} - i$.

- (i) Write down the other root of the equation. (1)

- (ii) Find the sum and product of the two roots of the equation. (3)

- (iii) Hence state the values of p and q . (2)

(Total 11 marks)

Q31.

The cubic equation

$$x^3 + px^2 + qx + r = 0$$

where p , q and r are real, has roots α , β and γ .

- (a) Given that

$$\alpha + \beta + \gamma = 4 \quad \text{and} \quad \alpha^2 + \beta^2 + \gamma^2 = 20$$

find the values of p and q .

(5)

- (b) Given further that one root is $3 + i$, find the value of r .

(5)

(Total 10 marks)

Q32.

The quadratic equation

$$3x^2 - 6x + 2 = 0$$

has roots α and β .

- (a) Write down the numerical values of $\alpha + \beta$ and $\alpha\beta$. (2)
- (b) (i) Expand $(\alpha + \beta)^3$. (1)
- (ii) Show that $\alpha^3 + \beta^3 = 4$. (3)
- (c) Find a quadratic equation with roots α^3 and β^3 , giving your answer in the form $px^2 + qx + r = 0$, where p , q and r are integers. (3)
- (Total 9 marks)

Q33.

The cubic equation

$$z^3 - 4iz^2 + qz - (4 - 2i) = 0$$

where q is a complex number, has roots α , β and γ .

- (a) Write down the value of:
- (i) $\alpha + \beta + \gamma$; (1)
- (ii) $\alpha\beta\gamma$. (1)
- (b) Given that $\alpha = \beta + \gamma$, show that:
- (i) $\alpha = 2i$; (1)
- (ii) $\beta\gamma = -(1 + 2i)$; (2)
- (iii) $q = -(5 + 2i)$. (3)
- (c) Show that β and γ are the roots of the equation
- $$z^2 - 2iz - (1 + 2i) = 0$$
- (2)
- (d) Given that β is real, find β and γ . (3)
- (Total 13 marks)

Q34.

The equation

$$x^2 - 5x - 2 = 0$$

has roots α and β .

- (a) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (b) Find the value of $\alpha^2\beta + \alpha\beta^2$.

(2)

- (c) Find a quadratic equation which has roots

$$\alpha^2\beta \text{ and } \alpha\beta^2$$

(3)

(Total 7 marks)

Q35.

The equation

$$x^2 - 4x + 13 = 0$$

has roots α and β .

- (a) (i) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

(2)

- (ii) Deduce that $\alpha^2 + \beta^2 = -10$.

(2)

- (iii) Explain why the statement $\alpha^2 + \beta^2 = -10$ implies that α and β cannot both be real.

(2)

- (b) Find in the form $p + iq$ the values of:

(i) $(\alpha + i) + (\beta + i)$;

(1)

(ii) $(\alpha + i)(\beta + i)$.

(2)

- (c) Hence find a quadratic equation with roots $(\alpha + i)$ and $(\beta + i)$.

(2)

(Total 11 marks)